Project Title: Project FloraForm: A Hybrid Generative AI & Robotics Framework for Autonomous Myco-Horticulture

Principal Investigator: Sabrina Hernandez Cavasos, Senior Lecturer, South Land University
Project Collaborator: Amanda Martin, PhD Student**,** Western Land University
Project Collaborator: Xiling Wu, Assistant Professor, University of North Land

NVIDIA Contact: Mahavir Jain, mahjn@nvidia.com
Dr. Cavasos and Dr. Jain were co-panelists at GTC 2024 ("Digital Twins for Next-Gen Agriculture"), and Dr. Wu was an NVIDIA intern with the Isaac Gym team in 2022.

Abstract
This project addresses a grand challenge in autonomous systems: predictive, real-time interaction with complex biological processes. We propose a hybrid "cloud-to-edge" system to autonomously cultivate the "Bioluminescent Myco-Truffle," an organism requiring precise, predictive care. We will first train a large-scale generative AI model in the cloud to predict fungal growth. This model will then be deployed locally on NVIDIA hardware to power a real-time digital twin and a physical robotic manipulator.

Project Keywords - Generative AI, Digital Twin, Robotics, Autonomous Horticulture

NVIDIA Platforms
This project is built entirely around the NVIDIA ecosystem, leveraging a hybrid cloud/edge architecture.

- Hardware Request: We will rely on A100 GPU hours for cloud training, an RTX PRO 6000 for the local digital twin, and Jetson AGX Orin Developer Kits for the physical robot.
- Software (Cloud): We have extensive experience with PyTorch and multi-node training. We will leverage NVIDIA's NGC containers for a reproducible environment to train our generative model on our 8-GPU node.
- Software (Local Server): The local RTX 6000 will run **NVIDIA Omniverse** to host the digital twin. The distilled "FloraForm" model will be served via the **NVIDIA Triton** Inference Server to ensure the lowest possible latency for digital twin synchronization.
- Software (Edge/Robot): The core simulation dataset was generated using **NVIDIA Isaac Sim**. The robot's manipulation and navigation policies will be trained in **NVIDIA Isaac Lab**. The Jetson Orin kits will run the full **NVIDIA JetPack** SDK, using the **TensorRT**-optimized model for real-time navigation and computer vision tasks. We will also use **RAPIDS** (**cuDF**) on the server to accelerate the analysis of incoming sensor-stream data.

Dataset and Model

- 25TB
- Synthetic hyperspectral video, simulated LiDAR, and time-series sensor data (e.g., humidity, soil ph, nutrient concentration).
- 100% generated in-house. This dataset is complete and ready for use.
- Data is non-confidential and contains no personal information. We plan to open-source a 5TB subset.
- Data is owned by our lab and is cleared for cloud use.
- The "FloraForm" generative model will have ~8 billion parameters. The distilled edge model will be ~1B parameters.
- Data is currently stored on our university's secure server (South Land University, USA), ready for upload to the cloud.

Introduction

The primary limitation of modern robotics is its inability to predict and reason about complex, non-rigid dynamic systems. While we can train a robot to pick up a box, we cannot train it to grow a plant, which requires anticipating the plant's needs days in advance. This project tackles this challenge by creating a "generative physical AI" that can model and predict complex biological growth.

Our test case is the fictional "Bioluminescent Myco-Truffle," which we have designed in simulation to be an "adversarial" organism—its health will collapse if 12 distinct environmental factors are not perfectly balanced. Our solution is Project FloraForm, a hybrid system that (1) learns a generative model of the truffle's life cycle in the cloud, (2) runs a predictive digital twin on a local server, and (3) acts on those predictions using a real-world robotic manipulator.

As shown in Figure 1, the cloud-trained model informs the Omniverse digital twin, which in turn directs the Jetson-powered robot. This "predict-and-act" loop, running in real-time, is the key innovation and is only possible with a hybrid compute architecture. Success will provide a blueprint for all autonomous science, from pharmaceutical research to advanced manufacturing.

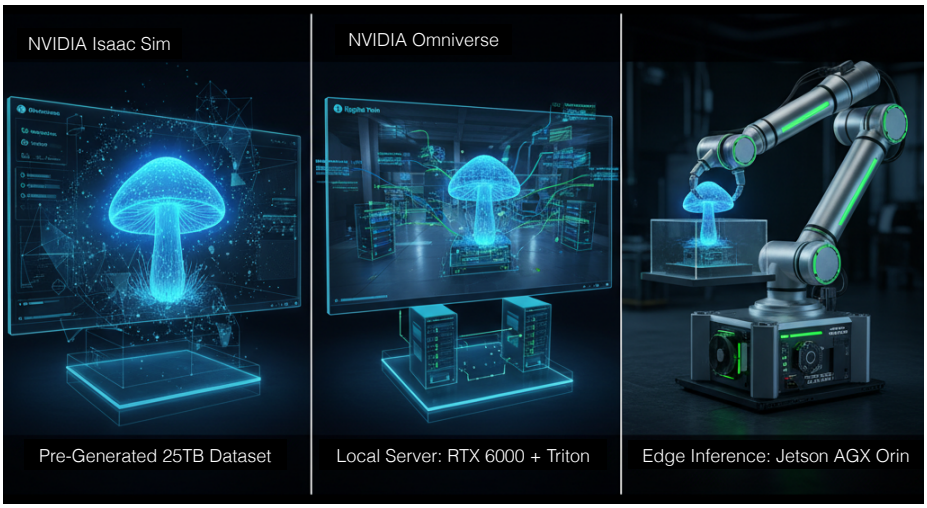


Figure 1: The Project FloraForm Concept

Methods

Our technical approach is broken into three phases. Our 25TB simulation dataset is already complete, allowing us to begin Phase 1 immediately upon receiving cloud access.

Phase 1: Foundation Model Training (Cloud): We will use the requested 32,000 A100 hours over the 6-month access window to train "FloraForm," our 8B parameter spatio-temporal transformer. This model will be trained on our 25TB dataset to predict the next 72 hours of truffle health based on the previous 24 hours of sensor data. This 6-month period is essential for conducting a large-scale hyperparameter sweep and training multiple model variants concurrently to achieve optimal predictive accuracy.

Phase 2: Digital Twin Deployment (Local Server): The 8B model is too large for real-time inference. We will use NVIDIA TensorRT to distill and quantize this model to a 1B parameter version. This model will be deployed on the local RTX 6000 Ada server using the NVIDIA Triton Inference Server. This server will run the Omniverse digital twin, ingesting data from 1,000+ real-world sensors, and feeding it to the local model.

Phase 3: Embodied AI (Edge): We will use Isaac Lab to train the control policies for a mobile manipulator. The robot, powered by two Jetson AGX Orin kits (one for navigation/CV, one for manipulation), will execute tasks (e.g., "adjust nutrient-valve-3," "prune-site-A"). The Jetsons are essential for on-board obstacle avoidance and real-time arm control, while the high-level "what-to-do" command comes from the local RTX 6000 server.

Proposed Timeline

Month/Year	Activity + Software, Models, Datasets Used	Compute Resource(s) Used
Jan – Feb 2027	• **Ingestion**: Upload pre-generated 25TB dataset to cloud storage. • **Training**: Launch architectural search and begin continuous training of the 8B "FloraForm" Foundation Model using PyTorch.	10K A100 hours [Cloud access begins]
Feb – April 2027	• **Setup and install RTX** • **Digital Twin:** Establish the real-time Omniverse digital twin on the local server; calibrate with lab sensors.	RTX 6000 PRO [expected arrival: February]
Mar – Apr 2027	**Convergence:** Continuous high-utilization training of the 8B model to reach convergence.	12K A100 hours
May – June 2027	• **Distillation**: Use cloud compute to distill the frozen 8B model into a 1B "Student" model for edge deployment. • **Egress**: Download final models and logs.	10K A100 GPU hours [Cloud access ends]
July – Aug 2027	• **Edge**: Finalize robot navigation policies in Isaac Lab.	Jetson Orin Kits
Sep – Nov 2027	• **Deployment**: Deploy final models to local hardware. • **Demo**: Conduct full autonomous cultivation trials. • Prepare submission for CVPR	Local hardware
Nov – Feb 2027	Prepare submission for IROS	N/A
January 2028	**Release**: Open-source code and models via GitHub.	N/A

Expected Results

1. Two conference publications (CVPR 2028 and IROS 2028).
2. An open-source release of the "FloraForm-Small" 1B parameter model via GitHub.
3. A 5TB "Myco-Truffle" dataset subset via our university's data portal.

Dependencies

The project does not depend on other grants or donations. The data is already collected and ready to be used.

Project Support Details

- **32,000 A100 hours in the cloud**:

    - Justification: Our request for the maximum 32,000 hours is based on the need for continuous, high-utilization training over the 6-month (approx. 35,000 total hours) access window. Training a single 8B parameter model is only the baseline. Our core research involves a large-scale, 6-month hyperparameter sweep and comparative training of several model architectures in parallel.
    - Calculation: We will run 4 concurrent experiments on the 8-GPU node (e.g., 4x 2-GPU jobs or 2x 4-GPU jobs) to explore different model backbones and optimization parameters. This parallel workload is essential for our research goals and will keep the 8-GPU node at ~92% utilization for the entire 6-month period, consuming the full 32,000-hour grant.
- **8 concurrent GPUs**: Required for our parallel experimentation and distributed training plan (PyTorch DDP).
- **32 TB of cloud storage**: To store our entire 25TB dataset, plus 7TB for model checkpoints, logs, and container images.
- **Physical Hardware**:
    - **1x RTX PRO 6000**: Cloud compute cannot be used for this task. The digital twin must synchronize with 1,000+ local lab sensors and provide predictions to the robot with sub-10ms latency. Cloud round-trip latency would make this impossible. The RTX 6000's 96GB of VRAM is the minimum required to run the 1B distilled model and the complex Omniverse scene simultaneously.
    - **2x Jetson AGX Orin Developer Kits**: Cloud compute is not feasible for on-board robotics. These are for the physical robot. We require one Jetson for the mobile base (real-time SLAM, CV) and one for the manipulator arm (real-time pose estimation, safety-critical grasping).

Cloud Readiness

Our team is fully prepared to begin training on day one of the access window.

- OS: Our development environment is Ubuntu 22.04.
- Activities:
    - Importing data: Our 25TB dataset is complete, validated, and staged on our university's server. We will use rclone to perform a one-time upload to the cloud storage in the first 24 hours.
    - Dependencies/Containers: Our entire workflow is containerized using Docker. We have a production-ready Dockerfile that builds from the NVIDIA PyTorch NGC container and adds our requirements.txt.
    - Executing code: We will git pull our code and launch our suite of parallel training jobs via a bash script.
    - Backing up: All model checkpoints and logs are backed up automatically to a Weights & Biases (wandb) project.